Water's bonds as a circuit

Does kronspice's AC analysis of a mass–spring circuit give the infrared fundamentals of H_2O, and predict its isotopologues? — doc/chemistry-plan.md

kronspice simulation report · 2026-11-07

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Summary

The vibration spectrum of a water molecule is computed by kronspice's AC analysis: atoms are mass capacitors, bonds are spring inductors, and Wilson's B matrix is an E/F (controlled-source) transformer between the bond frame and the atom frame. Three force constants are fitted to H_2O's three measured fundamentals; the same constants then predict D_2O, HDO and $H_2{}^{18}O$ with no refit, since isotopes change masses, not bonds. This is the first molecule of the chemistry plan.

- **The circuit is the molecule's equations:** its AC peaks equal the GF eigenvalues (Wilson's method) to 10^{-6}, for every mode of every isotopologue.
- **Fitted force field** (to the H_2O fundamentals): f_r 7.6712, f_{rr} −0.0942, f_a 0.7030 mdyn/Å; reproduces 1595, 3657 and 3756 cm^{-1} exactly.
- **D_2O is predicted within 1.32 %** (−1.32 %, −0.89 %, −1.29 %), **HDO within 1.09 %** (−0.48 %, −1.09 %, +0.05 %).
- **The remaining error is anharmonicity:** a harmonic field fitted to H_2O's fundamentals comes out soft and under-predicts the deuterium modes by about 1 %; the O–H stretch of HDO is within 0.1 %.
- **$H_2{}^{18}O$** (1589.15, 3647.51, 3740.70 cm^{-1}) is a prediction only: no verified measurement has been checked yet.

The construction

The mobility (force–current) analogy of Firestone 1933, the one Kron and Branin use for network analogies (CLAUDE.md §3a.1):

Mechanics	Circuit
velocity of a coordinate	node voltage
force	current
mass m	capacitor C = m to ground
spring of stiffness k	inductor L = 1/k
damper	resistor to ground

- The six in-plane Cartesian coordinates of O, H1 and H2 are nodes with the atomic mass as a capacitor.
- The three bond coordinates (two O–H stretches and the H–O–H bend) are springs. Each spring's rate of change is a linear combination of the atomic velocities, $s' = \Sigma\, B_{kj}\, v_j$ (Wilson's B matrix), built from a chain of voltage-controlled voltage sources (E); the spring force flows in an inductor; and that force acts back on every atom through current-controlled current sources (F). The E/F pair is an ideal multiport transformer, Kron's connection tensor between the two frames.
- The force constants form the inductance matrix $L = F^{-1}$, so the stretch–stretch interaction f_{rr} is a mutual inductance (K element).
- Circuit units: capacitance in daltons, inductance in Å/mdyn, so the circuit's $\omega = 1/\sqrt{(LC)}$ maps to the molecule's through $\sqrt{(100\ \text{N/m} / 1\ \text{u})}$; the netlist header gives f_c [Hz] × 8185.679725 = wavenumber [cm^{-1}].
- Geometry r_e = 0.958 Å, H–O–H angle 104.4776° (Hoy & Bunker 1979). Three force constants (f_r, f_{rr}, f_a; no stretch–bend term) are fitted by Newton's method to H_2O's three measured fundamentals.

An excerpt of *netlists/h2o.cir*, the first O–H stretch (masses in u, a large damping resistor on each node, the B-matrix chain of E sources, the spring inductor and the F sources that return its force):

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```
CM_ox ox 0 15.9949146196
RD_ox ox 0 1000000
E_r1_ox r1_s0 0 ox 0 -0.790569884316532
E_r1_oy r1_s1 r1_s0 oy 0 -0.612371829864621
E_r1_h1x r1_s2 r1_s1 h1x 0 0.790569884316532
E_r1_h1y r1_s3 r1_s2 h1y 0 0.612371829864621
VS_r1 r1_s3 r1_l 0
L_r1 r1_l 0 0.130378191754178
F_r1_ox ox 0 VS_r1 -0.790569884316532
...
K_rr L_r1 L_r2 0.0122798366009168
I_fx 0 h1x AC 1
I_fy 0 h1y AC 1
.ac lin 4001 0.0610822823 0.549740541
```

Results

Force fields

Fitted to	f_r [mdyn/Å]	f_{rr} [mdyn/Å]	f_a [mdyn/Å]
H_2O fundamentals (used)	7.6712	−0.0942	0.7030
H_2O harmonic frequencies	8.4388	−0.1191	0.7514

Vibration frequencies

The circuit's AC peaks (coarse sweep, fine sweep, parabola through the top three points) against the GF eigenvalues and the measured gas-phase fundamentals (NIST CCCBDB: Shimanouchi; Huber & Herzberg):

Molecule	Mode	Circuit AC peak [cm^{-1}]	GF eigenvalue	Measured	Error
H_2O	bend	1595.00	1595.00	1595.0	−0.00 % (fitted)
H_2O	stretch	3657.00	3657.00	3657.0	−0.00 % (fitted)
H_2O	stretch	3756.00	3756.00	3756.0	−0.00 % (fitted)
D_2O	bend	1162.51	1162.51	1178.0	−1.32 %
D_2O	stretch	2647.26	2647.26	2671.0	−0.89 %
D_2O	stretch	2752.05	2752.05	2788.0	−1.29 %
HDO	bend	1395.34	1395.34	1402.0	−0.48 %
HDO	stretch	2697.36	2697.36	2727.0	−1.09 %
HDO	stretch	3708.98	3708.98	3707.0	+0.05 %
$H_2{}^{18}O$	bend	1589.15	1589.15	—	prediction only
$H_2{}^{18}O$	stretch	3647.51	3647.51	—	prediction only
$H_2{}^{18}O$	stretch	3740.70	3740.70	—	prediction only

- H_2O is fitted (0 % by construction); D_2O, HDO and $H_2{}^{18}O$ are predictions with the same bonds.
- $H_2{}^{18}O$ is not in CCCBDB and its band origins have not been checked against a primary source, so it is shown as a prediction.
- The remaining error is anharmonicity. A measured fundamental sits below the harmonic frequency by a shift that shrinks with heavier atoms (ω·x scales as 1/μ against ω as 1/√μ, with μ the reduced mass and x the anharmonicity constant). A harmonic force field fitted to H_2O's fundamentals absorbs hydrogen's larger shift, so it comes out soft and under-predicts the deuterium modes by about 1 %.

Validation

- AC peaks equal the GF eigenvalues to 10^{-6}: the circuit is the molecule's equations, not an approximation of them.
- Tests (*main_test.go*): the fit reproduces H_2O; AC peaks equal eigenvalues; isotopes within 1.5 %; the three zero modes (two translations, one in-plane rotation) are present; the force field is physically sensible (stretch stiffer than bend, a weak negative stretch–stretch interaction).

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- Each isotopologue is a plain SPICE netlist (E, F, K elements) in *netlists/*, runnable in kronspice or any SPICE; *export/* holds the same netlists with KiCad schematics and results sheets.

Limits

- Harmonic springs only: anharmonicity is the remaining ~1 % error. The next step in the plan is Morse (nonlinear) inductors.
- No stretch–bend coupling term; three constants for three H_2O lines.
- In-plane motion only; gas-phase single molecule.

Reproducing the results

```
go run ./examples/water_vibration [-netlists dir]
go test ./examples/water_vibration
```

Sources: F. A. Firestone, J. Acoust. Soc. Am. 4, 249 (1933); E. B. Wilson, J. C. Decius, P. C. Cross, Molecular Vibrations (1955), Ch. 4; A. R. Hoy, P. R. Bunker, J. Mol. Spectrosc. 74, 1 (1979); NIST CCCBDB (T. Shimanouchi; K. P. Huber, G. Herzberg, 1979); NIST "Atomic Weights and Isotopic Compositions".

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